MIT SCHOOL OF COMPUTING
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MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject-PL.

Q.1.) (i) List:- ordered and mutable
(ii) topic:- ordered and immutable
(iii) String:- ordered and immutable
(iv) Dictionary:- Ordered and mutable
(v) Set:- unordered & Muttable, Store
unique element only.

2) Indexing:- It is used to access a
single element using it position (start from 0)

Slicing:- It is used to access a part of
the sequence using [Start : End : STEP]
Stop

Ex:- S = 'python'.
L = [10, 20, 30, 40, 50]

Indexing:- S[2] = 't'
L [2] = 30.

∴ Slicing ⇒ S[0:6:2] = 'Pto'
L[0:6:3] = [10,40]

* Difference (1) String cannot be modified after slicing (immutable (2) List can be modified after slicing (mutable)

Q3) Ch = 'Hello World'
print("First Character:", ch[0])
print("Few Letters:", ch[0:6:2])
print("Upper case format:" ch.Upper())
print("Lower Case Format:", ch.Lower())

Output :- First character: 'H'
Few Letters: 'Hlo'
Upper case format: ~~'Hellow World~~ 'HELLOW WORLD'
Lower case format: 'helloworld'

Q4) Equality (==) checks whether values are same or Not. Identity (is) checks whether both variable points to Same memory location

Ex- a = [1, 2, 3]
b = [1, 2, 3]
print (a==b)
print (a is b)

∴ Explanation
(i) List are mutable and store in different memory Location
(ii) (==) Compares value
(iii) (is) compares memory reference.

Output :-
True
False

Q.5) List:- Allows duplicates, slower searching
Set:- Stores only unique values, no indexing
Dictionary:- Stores unique student IDs as keys and detail in value.

∴ Most Efficient- Dictionary, because it allow fast access, unique keys and structured data storage.

Q.6) St_Record = {'100':'Rahul', 101:'Ram', 102:'Sia'}
print ("student name of ID No.100:", St_Record[0])
print ("Total No. of students:" len (St_Record))
Print ("St_Record keys in List:" list(St_Record.keys()))

Output:- Student name of ID No.100: 'Rahul'
Total No. of students: 3
St_Record keys in List: [100, 101, 102]

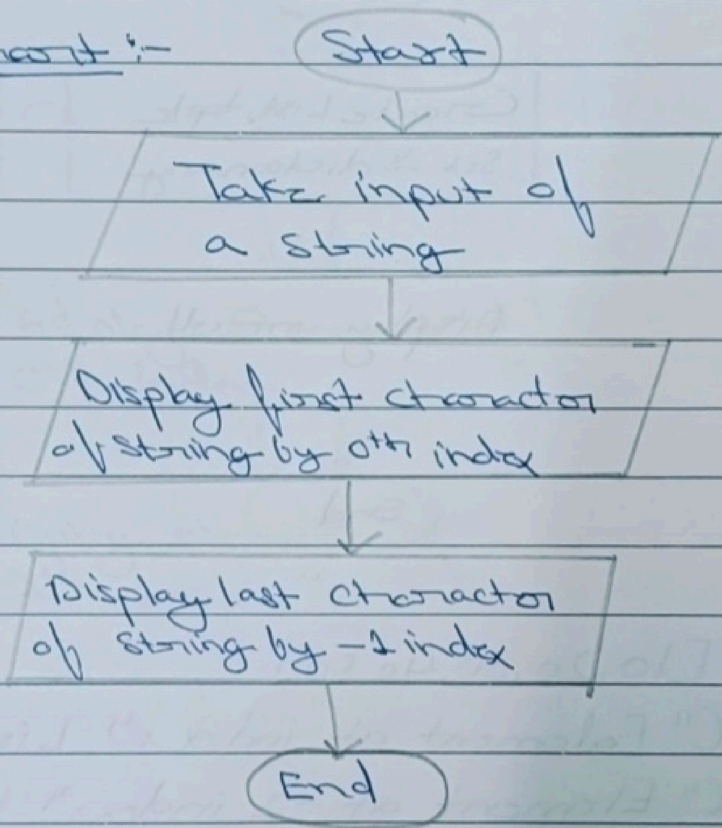
Q7 Str = input ("Enter a String:")
print ("First charactor of string:", str[0])
print ("Last charactor of string:", str[-1])
Enter a String: Hello

Output: First charactor of String: 'H'
Last charactor of String: 'O'

* Algorithm

1) Start
2) Take input of a string
3) Display first character of string by 0th index
4) Display last character of string by -1th index
5) End

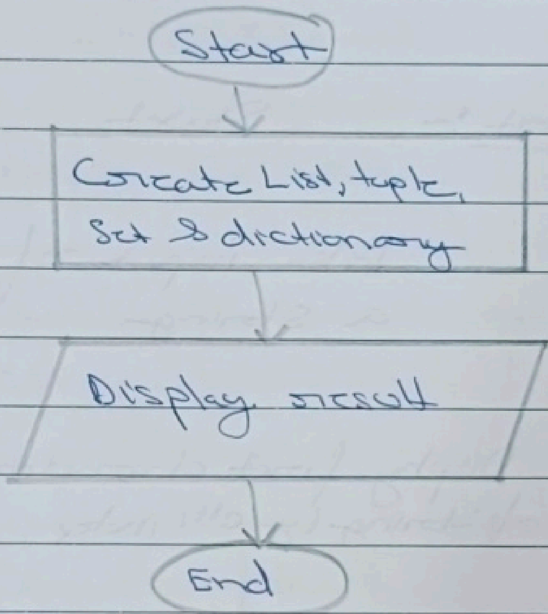
* Flowchart :-



Q8
```
List= [1,2,3, "apple"]
my_tuple = (10,20,30)
Set_from_list = set([1,2,3,4])
my_dict = {"name": "Animal", "age":12}
print(list, my_tuple, set_from_list, my_dict)
```

* Algorithm
1) Start
2) Create List, tuple, Set and Dictionary
3) Display result
4) Stop/End

* Flowchart:-

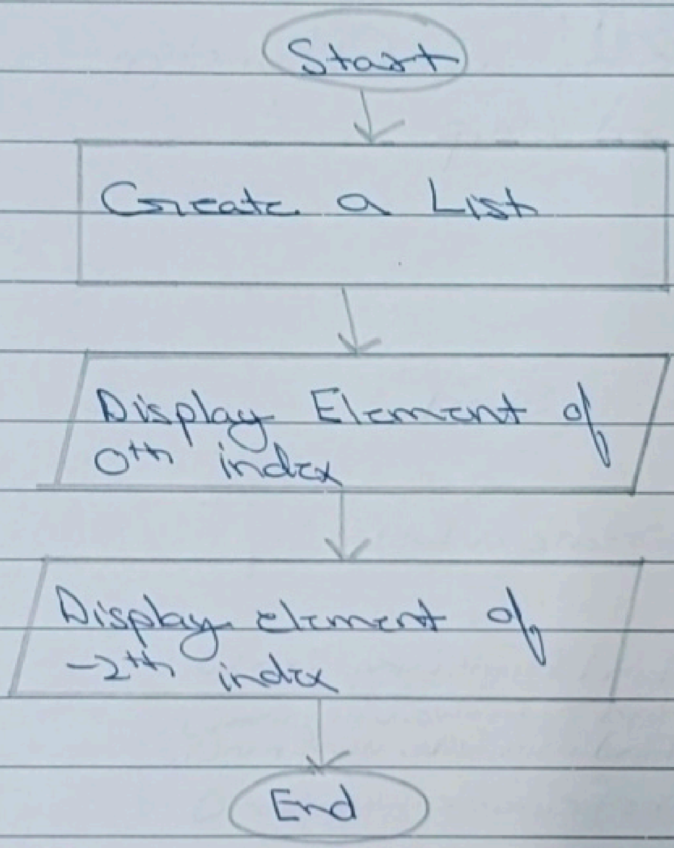


Q.9) List = [10,20,30,40,50]
print ("Element at index :", List [0])
print ("Element at -2 index:" List [-2])

Output :- Element at 0 index: 10
Element at -2 index: 40

* **Algorithm:-** (1) Start
(2) Create a List
(3) Display element of 0^{th} index
(4) Display element of -2^{th} index
(5) Stop/End

* **Flowchart:-**



Q10)
```
List = [5,2,9,1]
Print ("Length:", len(list))
print ("Maximum:" max(list))
print("Minimum:" min (list))
print("Number in descending order:", list.Sort(reverse=True)
```

Output:- Length: 4
Maximum: 9